

SARDAR PATEL UNIVERSITY**M. Sc. I. T. (Integrated) Examination, 7th Semester****Monday, 13th November, 2017****PS07EIT01: Image Processing****Time: 02:00 PM to 04:00 PM****Total Marks: 70****Q:1 Give answers of following Multiple Choice Questions****[08]****[01]** An image is considered to be a function of $a(x, y)$ where a represents

- (A) height of image (B) width of image
(C) amplitude of image (D) resolution of image

[02] 2D Fourier transform and its inverse are infinitely

- (A) Periodic (B) Aperiodic
(C) Linear (D) Non-linear

[03] Which one is not the source of images?

- (A) Mecatronic (B) Acoustic
(C) Ultrasonic (D) electronic

[04] Square of standard deviation is called

- (A) Noise (B) Restoration
(C) Variance (D) Power

[05] Images usually gets corrupted during

- (A) Transmission (B) Degradation
(C) Restoration (D) Acquisition

[06] Impulse is simulated by

- (A) Gray dot (B) Black dot
(C) Bright dot (D) Sharp dot

[07] Encoder is used for

- (A) Image enhancement (B) Image decompression
(C) Image equalization (D) Image compression

[08] Morphology refers to

- (A) Pixels (B) Pixels
(C) Matrix (D) Shape

Q:2 Answer the following short questions (any Seven)**[14]****[01]** What is Digital Image Processing?**[02]** What is image sensors?**[03]** What is image transform? Explain in brief.**[04]** List out smoothing techniques.**[05]** Write the equation of Fourier and Inverse Fourier transform.**[06]** What is image degradation and restoration?**[07]** What is the disadvantage of wiener filtering?**[08]** Which are the data compression techniques?**[09]** List-out techniques of Bit-Plane coding.**P. T. O.**

- Q:3 [A] Explain Elements of visual perception. [06]
[B] Explain any three components of an image processing system. [06]

OR

- [B] Explain any three applications of Image Processing [06]

- Q:4 [A] Explain Fourier Transform and Inverse Fourier Transform. [06]
[B] Explain any two properties of the 2-D Discrete Fourier Transform. [06]

OR

- [B] Explain any two Image Sharpening techniques. [06]

- Q:5 [A] Explain Noise Probability Density Functions. [06]
[B] Explain a model of the Image degradation / restoration process. [06]

OR

- [B] Explain Estimation of Noise Parameters. [06]

- Q:6 [A] Explain source encoder & decoder. [06]
[B] Explain measuring information and information channel [06]

OR

- [B] Explain Bit-plane coding. [06]

— X —
②